MILITARY SPECIFICATION

ATTACHMENTS, CUTTING FOR TORCHES, OXYACETYLENE WELDING AND CUTTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers replacement cutting attachments for hand held oxyacetylene torches used for cutting ferrous metals.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D572-81 - Standard Test Method for Rubber-Deterioration by Heat and Oxygen
ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA E-5 - Torch Standard for Welding and Cutting

(Requests for copies should be addressed to the Compressed Gas Association, Inc., 1235 Jefferson Davis Hwy, Arlington, VA 22202.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The cutting attachment shall be a replacement for the brand name and model number cutting attachment as specified in the contract (see 6.2.1), and shall meet the requirements of CGA E-5, of the positive medium pressure type, incorporating facilities for mixing and controlling the flow of gases to the torch cutting tip (see 6.5). Positive medium pressure is pressure from one to fifteen pounds per square inch gauge (psig). The cutting attachments shall fit the brand name and model number of the torch handle and cutting tip specified in the contract or purchase order and shall meet all the requirements specified herein.

3.2.1 Construction. All parts used in fabrication of the cutting attachments shall be new, clean, and free of oil and grease. In no event shall patching processes be used for reclaiming defective parts. Surfaces of all parts shall be free of burrs, defects, and dirt. Parts subject to removal or adjustment shall not be permanently deformed by the assembly process.

3.2.1.1 Threads. Screw threads shall comply with the requirements of FED-STD-H28 and the Detailed Standard sections referenced therein. All internal and external threads shall be clean, free of any extraneous materials, and shall have no burrs or nicks.

3.2.1.2 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the attachments.

3.2.1.3 Brazing. Brazing shall be of the highest quality workmanship prevailing among commercial manufacturers normally producing cutting attachments of the types specified herein. Flux shall be cleaned from surfaces after brazing is completed. Brazing filler metal shall have a melting temperature no lower than 1000 degrees Fahrenheit (F). Filler metal shall be visibly complete around all brazed joints.

3.2.1.4 Machined surfaces. Gas holes and passages shall be free of burrs. Machined surface seals shall be free of nicks, burrs, and scratches and shall meet the leakage requirements of 3.3.5.

3.2.1.5 Lubricants. Lubricants used inside the cutting attachment shall be compatible with oxygen at 90 psig.

3.2.1.6 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free of defects that adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

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3.3 Performance and product characteristics.

3.3.1 Capacities. Cutting attachment flow rate capacities shall conform to the requirements specified in 3.3.2 and 3.3.3 with the brand name cutting tip of the following size installed.

4 or 6 Preheat holes Drill Size Number	Diameter (inches)	Cutting Oxygen hole Drill Size Number	Diameter (inches)
53 to 69	.0595 to .0292	43 to 31	.089 to .120

3.3.2 Oxygen flow rate. The cutting attachment, when attached to the handle specified in the contract, with the above specified cutting tip installed, shall have a cutting oxygen flow rate of 450 standard cubic feet per hour (scfh) or more at a torch oxygen inlet pressure not exceeding 90 psig.

3.3.3 Acetylene flow rate. The cutting attachment, when attached to the handle specified in the contract, with the above specified cutting tip installed and the preheat flame adjusted to neutral, shall have an acetylene flow capacity of at least 25 scfh at a torch acetylene inlet pressure not exceeding 10 psig.

3.3.4 Backfire and flashback. Each torch mix cutting attachment and each tip mix cutting attachment shall withstand five backfires (see 6.5), at a rate of at least one per second with no flashback (see 6.5), when the oxygen and acetylene are set at the manufacturer's recommended pressures and the preheat flame is adjusted to neutral. WARNING: A flashback condition necessitates immediately cutting off the gas supply (oxygen first, acetylene second) to prevent damage and possible destruction of the torch, gas tubes, or hoses.

3.3.5 Leakage. Joints, valves, valve packing, o-ring seals, metal to metal mating seals, gas passages, connections, and glands shall sustain gas pressures of at least 50 psig internally (from one gas passage to another) and externally (from internal gas passage to external atmosphere). Valves shall sustain the gas pressures without leakage that is sufficient for formation of a bubble when submerged in clean water for ten seconds, both when the valves are closed and when the valves are open to gas passage. The cutting tip seats shall not allow gas leakage at 50 psig, or higher, when tightened with no more than 20 foot-pounds of torque applied to the tip nut and the brand name and model series of the tip specified in the contract or purchase order is installed in the head.

3.4 Details of components. Cutting attachments shall consist of a mating part and retaining nut for mounting to the welding torch handle, gas tubes for carrying gases to the tip, a cutting torch head, a tip retaining nut, a preheat oxygen control valve, and a lever operated valve to control cutting oxygen. The preheat oxygen valve and the cutting oxygen valve are not required to be in the cutting attachment if those functions are provided in the welding torch handle. Cutting attachments shall be mountable onto the handle with a gas tight fit at 50 psig, conforming to 3.3.5, by hand tightening only and without the use of a tool.

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3.4.1 Preheat oxygen valve.

3.4.1.1 Stems. Valve stems shall be fabricated of corrosion-resistant metal. Valve stems shall maintain a set valve adjustment during operation of the torch. Stems of the oxygen valve shall be assembled to prevent removal from the valve without first removing the gland nut and packing.

3.4.1.2 Packing. The valve packing shall be made of material compatable with oxygen. The valve packing, or alternate means, shall provide a drag equivalent to at least four inch-ounces of torque on the valve stem to aid the stem in maintaining its setting during use, and to prevent the stem from being turned by accident.

3.4.1.3 Knobs. Valve knobs shall be fabricated of corrosion-resistant metal. Valve knobs shall be 0.875 inch in diameter or larger and shall provide a positive grip for an operator wearing welding gloves. Valve knobs shall not separate from the stem when pulled with a force of at least 70 pounds along the centerline of the stem. Valve knobs shall not loosen or separate from the stem when at least 50 inch-pounds of torque is applied to the knob in both the clockwise and counter-clockwise directions.

3.4.2 O-rings. O-rings shall not crack or show visible evidence of deterioration following exposure for 96 hours to oxygen, at a pressure of 300 psig and at a temperature of 158 degrees F, when tested in accordance with ASTM D572-81. O-rings shall not fail to provide adequate leak free seals, in accordance with 3.4.1.1, due to chemical attacks of the gases used, or because of normal operating and environmental conditions.

3.4.3 Gas tubes. Gas tubes in the cutting attachment shall not permanently deform, nor shall the brazed joints fail when a wrench is used to tighten the tip nut with no less than 40 foot-pounds of torque applied to the tip nut at 90 ±5 degrees to the tubes, and with the tubes supported no closer than five inches from the centerline of the tip nut. Gas tube joints shall be brazed to the cutting head. Brazing shall conform to 3.2.1.3.

3.4.4 Cutting head. The cutting attachment head shall be fabricated of copper alloy or stainless steel, and shall have a 90 ±2 degree angle of the tip position relative to the centerline of the welding torch handle. The cutting head shall incorporate a tip nut that shall hold the tip against the seat of the cutting head.

3.4.5 Cutting oxygen control valve. The cutting oxygen control valve shall not leak externally at 50 psig, or more, when open and when closed. The cutting oxygen control valve shall not leak through from the up-stream side of the valve to the down-stream side of the valve when subjected to 50 psig or more with the valve closed. All of these conditions shall be met after the cutting oxygen control valve has been opened and closed by the lever at least 1000 times. There shall be no bent, broken, or loose parts after at least 1000 cycles of operation.

3.4.5.1 Cutting oxygen valve stem. The cutting oxygen valve stem shall be mechanically retained in the valve by the lever such that the lever cannot release the stem without the lever being disassembled from the torch or the stem shall be internally retained such that it shall be securely held in the torch when the valve is pressurized from upstream with 150 psig air pressure and the lever rotated away from the valve stem.

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3.4.6 Lever. The cutting oxygen valve lever shall be fabricated of corrosion-resistant metal. Levers shall not permanently deform under 30 pounds or more of force, when applied toward the centerline of the torch handle at least four inches from the lever fulcrum or next nearest supporting point to the end of the lever, or within 1/2 inch of the end of the lever if less than four inches long.

3.5 Mixing. Gas mixing shall occur in the cutting attachment or the tip as specified in the contract. (See 6.2.1)

3.6 Product identification. The cutting attachments shall be marked for product identification in accordance with MIL-STD-130. Unless otherwise specified (see 6.2.1), all cutting attachments shall be permanently marked with the manufacturer's name and model number. Each cutting attachment shall be packed with a parts list including an exploded view of the cutting attachment and all field replaceable parts including o-rings, valve stem/knob assemblies and packings, and tip nut. All replaceable parts shall be identified with the original equipment manufacturer's part number. O-rings shall be identified with the o-ring manufacturer's name and part number as well as the torch manufacturer's part number.

3.7 Workmanship. Workmanship of cutting attachments shall be of the highest quality prevailing among manufacturer's normally producing cutting attachments of the type specified herein for the commercial market.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

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4.3 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on the preproduction sample(s) or first production item(s) specified in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examinations in 4.5, and all tests in 4.6. Failure of the first article to pass any examination or test shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (d) as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

a. Product examination (see 4.5)

b. Leak test (see 4.6.1)

c. Flame tests (see 4.6.3)

d. Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105 using inspection levels and acceptable quality levels (AQL) as listed unless 100 percent inspection is specified.

Inspection or Test	Inspection level	AQL
Product examination (see 4.5)	II	1.0
Leak test (see 4.6.1)	100 percent	-
Flame tests (see 4.6.3)	100 percent	-
Packaging inspection (see 4.7)	S-4	4.0

4.5 Product examination. Visually, dimensionally, and manually examine each cutting attachment to determine conformance with the requirements of 3.2 through 3.2.1.6, 3.4 through 3.4.4 and 3.5 through 3.7. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Dimensional examination includes measuring dimensions including weights, as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the unit and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so all replacement parts will not degrade the form, fit, or function of the end item (see 3.2.1.2), shall be cause for rejection.

4.6 Tests. Tests shall be conducted using pressure gauges calibrated to within three percent of full scale at all readings in the upper and lower 1/4 portions of the scale and shall be accurate within two percent within the center 1/2 portion of the scale. Flow meters shall be accurate within three percent of full scale reading. Failure of the cutting attachment to meet any of the test requirements herein shall be cause for rejection.

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4.6.1 Leak. Attach an air or oxygen supply hose to the oxygen inlet valve and close the valve by tightening the knob to no more than 20 inch-pounds of torque. Submerge the handle in clean water with the hose end down. Apply a minimum of 50 psig pressure to th control valve. No bubbles shall form or any other evidence of leakage occur while being observed for a continuous ten second period. The same test procedure shall be repeated for the acetylene inlet valve. Attach the specified brand name welding nozzle with the mixers to the handle, tightening by hand, and plug the outlet of the welding tip. Open and pressurize both valves to a minimum of 50 psig. There shall be no formation of bubbles or any other indication of leakage anywhere on the torch assembly, including the hose inlet threads. Any incurable leak shall be cause for rejection. Torque requirements stated for this test shall not br exceeded at anytime during the test. For first article testing only, the cutting oxygen valve shall be opened and closed by the cutting oxygen lever at least 1000 times prior to the leak tests.

4.6.2 Flow capacity. The cutting attachment shall be attached to test equipment consisting of the brand name and model number of the handle specified in the contract and regulated supplies of oxygen and acetylene, each with a flow meter and pressure gauge in series in their delivery lines. The pressure gauges shall be located next to the torch inlets between the torch check valve and the flow meters. A cutting tip of the size specified in 3.3.1, shall be inserted in the cutting head. The oxygen flow capacity with the cutting valve fully open shall be not less than 450 scfh with the inlet pressure set at 90 psig or less (see 3.3.2). The torch shall then be lit and adjusted for a neutral flame, with the cutting oxygen valve closed. The acetylene regulator shall be adjusted to produce a flow rate of not less than 25 scfh with the inlet pressure at 10 psig or less (see 3.3.3).

4.6.3 Flame. Each cutting attachment shall be tested for leaks at the cutting head and for flame stability, as follows:

a. With the specified cutting tip attached (see 3.3.1), the torch shall be lit and adjusted to produce a neutral flame at the gas pressure recommended for the tip by the manufacturer. Repeated popping or backfiring of the flame is indicative of an internal leak and shall be cause for rejection.

b. The cutting oxygen valve shall then be fully opened and the preheat flame observed for changes. Other than elongation, any distortion of the flame is an indication of an internal leak and shall be cause for rejection.

c. A pilot light or other equally sensitive method shall be used to check for leaks at the tip nut. Any leakage at the tip nut shall be cause for rejection.

d. For torch mix torch cutting attachments, the end shall be tapped against a fire brick, carbon block, or other hard surfaces to produce a succession of five backfires at the rate of at least one per second. Any flashback as indicated by a pale blue flame accompanied by a sustained high-pitched whistle and possibly sparks issuing from the flame holes and/or the handle becoming hot to the touch, shall be cause for rejection (see 3.3.4). WARNING: If a flashback does occur, the oxygen must be closed immediately, then the acetylene valve must be closed. If the gas valves are not closed quickly, permanent damage to the torch and/or injury to personnel may result.

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4.6.4 Cutting oxygen valve lever. The torch shall be secured in a vise or other suitable fixture so that the cutting attachment is in the horizontal position with the cutting oxygen valve lever on top. A point load of 30 pounds shall be applied straight downward on the center of the lever at a point at least 4 inches from the nearest supporting part or within one-half inch of the lever end if the lever is less than four inches long (lever fulcrum or valve stem). Any permanent bend in the lever caused by the application of the load shall be cause for rejection (see 3.5.4).

4.6.5 Throttling valve knob. For this test the stem with knob assembly may be removed from the cutting attachment to prevent damage. Secure the cutting attachment or the stem in a vice or other suitable fixture. Apply a force of at least 70 pounds to the knob, directly away from and in line with the stem. Apply a torque of at least 50 inch-pounds to the knob in the clockwise direction. Apply the same torque in the counter-clockwise direction. The knob shall not loosen on the stem (see 3.4.1.4).

4.6.6 Gas tube strength. The cutting attachment shall be secured in a vise or other suitable fixture so that the gas tubes are supported no closer than five inches from the centerline of the tip nut. The tip nut shall be tightened with a torque of no less than 40 foot-pounds applied at 90±5 degrees to the tubes. There shall be no permanent deformation or failure of the brazing (see 3.4.3).

4.6.7 Cutting oxygen valve stem. The cutting oxygen valve stem test shall consist of two parts. The first is the lever that operates the valve stem shall be visually examined and manually operated to determine whether or not the lever itself mechanically or physically retains the valve stem such that it cannot come out of the valve without first disassembling the lever. If the valve lever physically retains the valve stem and cannot be rotated away from the valve stem then the design passes the requirement. If the valve lever can be moved to uncover the valve stem then the valve shall be disassembled and examined to determine whether or not the design of the valve is such that the valve stem cannot pass completely through the hole in the valve or pass through far enough to allow leakage of oxygen. Second the valve shall be reassembled and pressurized from the upstream side to 150 psig or more with compressed air and with the lever moved away from the valve stem. Failure of the valve to maintain 150 psig air pressure without leakage when submerged in water with the lever moved away for a continuous period of ten seconds or failure of the valve stem to remain in the valve when pressurrized to 150 psig for a continuous 30 seconds in the atmosphere shall be cause for rejection (see 3.4.5.1).

4.7 Packaging inspection. The torches shall be inspected before and after packaging to determine compliance with Section 5 herein.

5 PACKAGING

5.1 Military preservation, packing and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A and B packing, and military marking shall be in accordance with MIL-W-45562.

5.2 Industrial preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall be in accordance with ASTM D3951-82.

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6 NOTES

6.1 Intended use. The attachments (torch assemblies) covered by this specification are intended for use by the military for field maintenance operations in support of a diverse representation of military equipment, including tactical wheeled and tracked vehicles and weapon systems. The attachments is intended for use with acetylene fuel gas and may frequently be used by personnel relatively inexperienced at welding and cutting. The equipment will most often be used in unsheltered areas, and may likely be stored in dusty or damp environments.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Brand name, type of cutting tip and and model number of cutting attachment being replaced (see 3.2).

c. First article, when required (see 3.1).

d. Type of mixing arrangement (see 3.5).

e. Product identification, if different (see 3.7).

f. First article inspection, if different (see 4.3).

g. Level of preservation, packing and marking, if different (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the attachments should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government is presently appropriate for the pending contract.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract. (See 3.8)

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6.5 Definitions.

6.5.1 Backfire. Backfire is the explosion of gases due to a momentary retrogression of the flame into the torch tip, usually caused by snuffing the burning tip against any surface or by an imbalance of gases. A typical backfire (a loud "pop") results in immediate, automatic extinguishing of the flame and, if an ignition source exists outside of the tip, the mixed gases then promptly relight and burn properly.

6.5.2 Flashback. Flashback is a retrogression of the flame into the torch as far back as the mixing chamber and a sustained burning of gases inside the torch or tip.

6.5.3 Positive pressure. Positive pressure is acetylene pressure from one to fifteen psig.

6.5.4 Standard cubic feet per hour. Standard air is defined in the Compressed Air and Gas Handbook, copyright 1973, by the Compressed Air and Gas Institute, as air at a temperature of 68 degrees Fahrenheit (F), a pressure of 14.70 pounds per square in absolute (psia) and a relative humidity of 36 percent (0.0750 density). Standard cubic feet per hour is the delivery rate of standard air.

6.6 Subject term (key word) listing.

Acetylene
Attachment, cutting
Backfire
Compressed Gas Association
Cutting attachment
Flashback
Oxygen
Welding apparatus

Custodians:

Army - AL
Navy - YD
Air Force - 99

Review activities:

Air Force - 84

Preparing activity:

Army - AL

Project No. 3433-0111

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-45997	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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